

Lecture-10

→ Recap of Cascode stage.

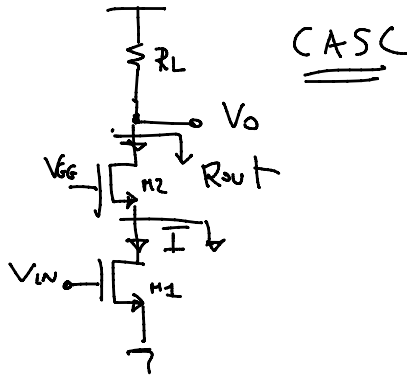
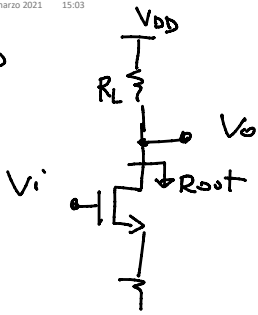
- To revise Cascode in a Go Follow the start of this lecture from (00:00 to 34:00)
- PMOS version of Cascode stage (34:00 — 46:00)
(46:00 — 60:00) Break
- Gain Boosted cascode (60:00 —

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EBIT, Unibo

Recap of cascode stage

cascode 2
lunedì 15 marzo 2021 15:03

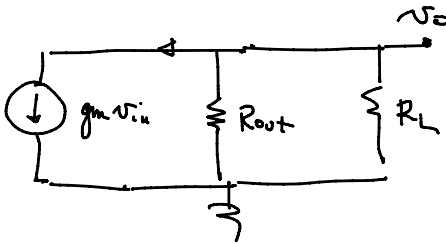
C.S



$$R_{out\,CS} = r_o \equiv v_{ds} \equiv \frac{1}{g_o}$$

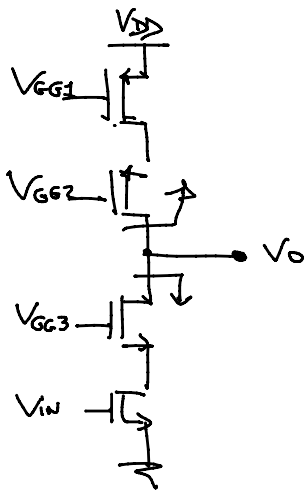
$$R_{out\,CASC} = \underbrace{g_{m2} \cdot v_{o2}}_{\sim 100} \cdot v_{o1}$$

V_{in}



$$\frac{v_o}{v_{in}} = -g_m (R_{out} \parallel R_L)$$

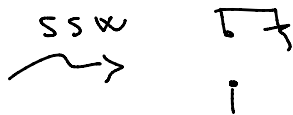
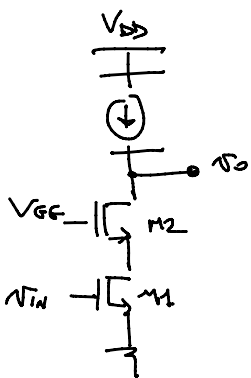
CS / CASC



$$R_1 \parallel R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$\text{if } R_1 \ll R_2 \Rightarrow R_1 \parallel R_2 \sim R_1$$

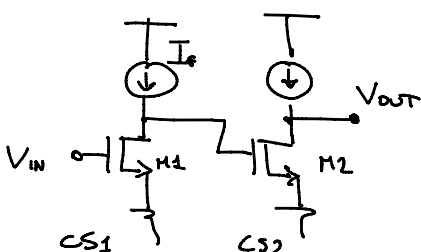
$$\frac{v_o}{v_{in}} = -g_m (R_{out\,CASC\,N} \parallel R_{out\,CASC\,P})$$



$$R_{out\,IDEAL\,CG} = \infty$$

$$\left[\frac{v_o}{v_{in}} = -g_{m1} \cdot g_{m2} \cdot v_{o2} \cdot v_{o1} \right]$$

Cascode of CS

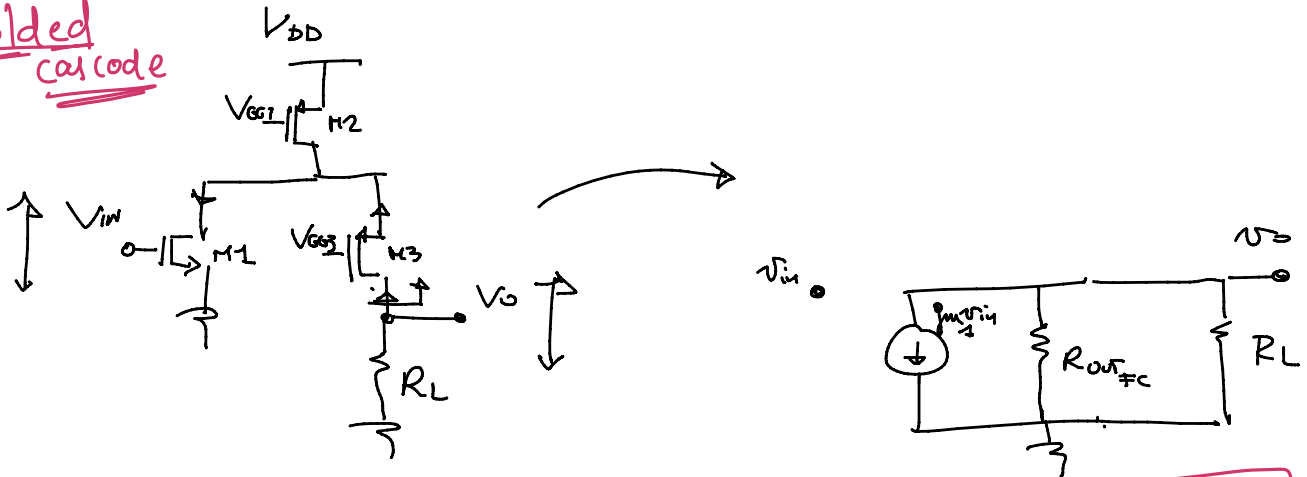


$$\left| \frac{v_o}{v_{in}} \right| = \left| g_{m1} \cdot v_{o1} \cdot g_{m2} \cdot v_{o2} \right|$$

Cascode of CS

	POWER	SWING	STABILITY
CASCODE	☺	☹	☺
CASCODE (or CS)	☹	☺	☹

Folded cascode



Gain of Folded cascode

$$\frac{v_o}{v_i} = -g_{m1} \cdot (R_{out,c} \parallel R_L)$$

• $M_{1,2,3} \dots$ ON/SAT

$$g_m \cdot r_o \gg 1$$

• ON

$$V_{GS} > V_{TN}$$

$$V_{GS} = V_{TN}$$

$$I_D = 0$$

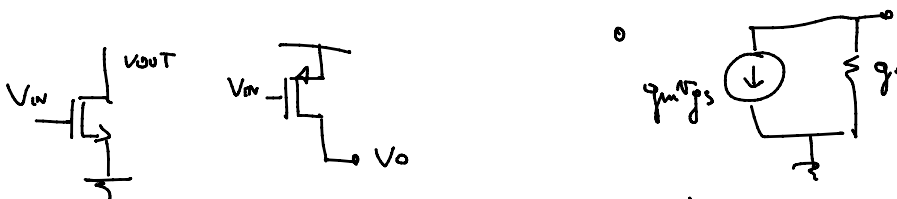
$$\hookrightarrow V_{GS} = V_{TN} + V_{OV} \rightarrow I_D$$

• SAT

$$V_{DS} \geq \underline{V_{OV}}$$

- All Transistors should be ON & in saturation
- Because only in saturation

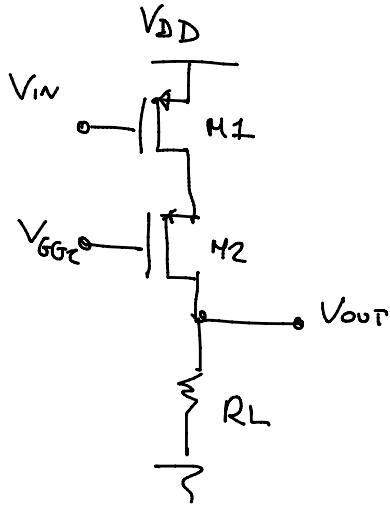
$$g_m \cdot r_o \gg 1$$



SSC

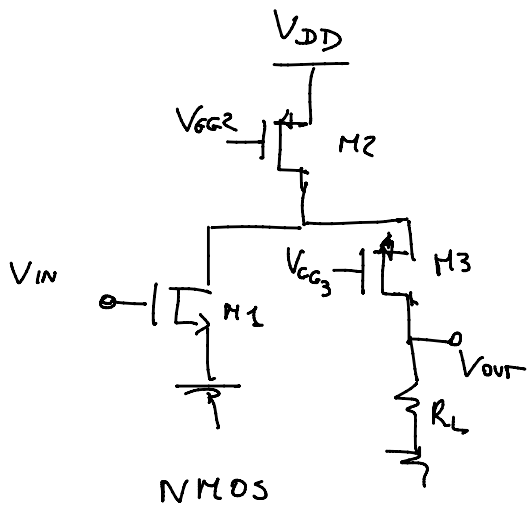
Recap from 00:00 to 34:00

PMOS Version of Cascode Stage



PMOS CASCODE

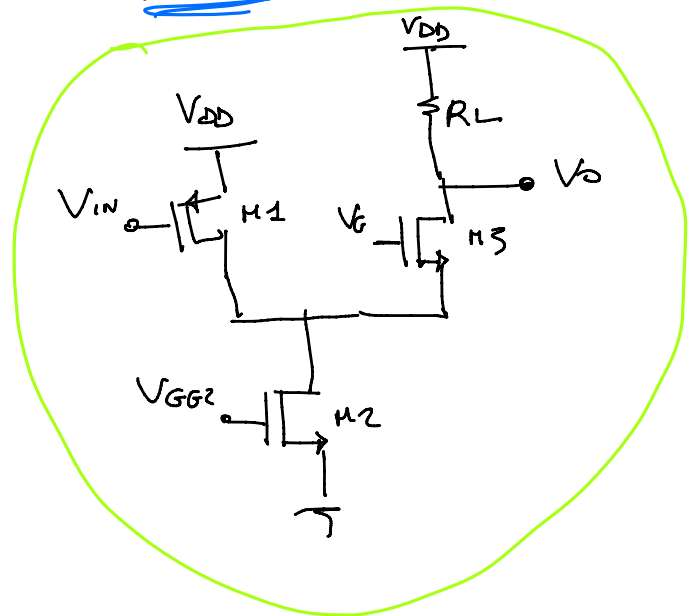
NMOS cascode



NMOS

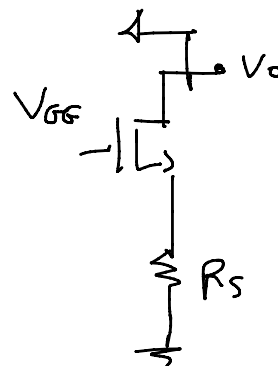
→ PMOS

PMOS cascode



$$g_m \cdot (R_{out_c} \parallel R_L)$$

$$R_{out_{casc}} \approx r_o \cdot g_m \cdot R_S$$

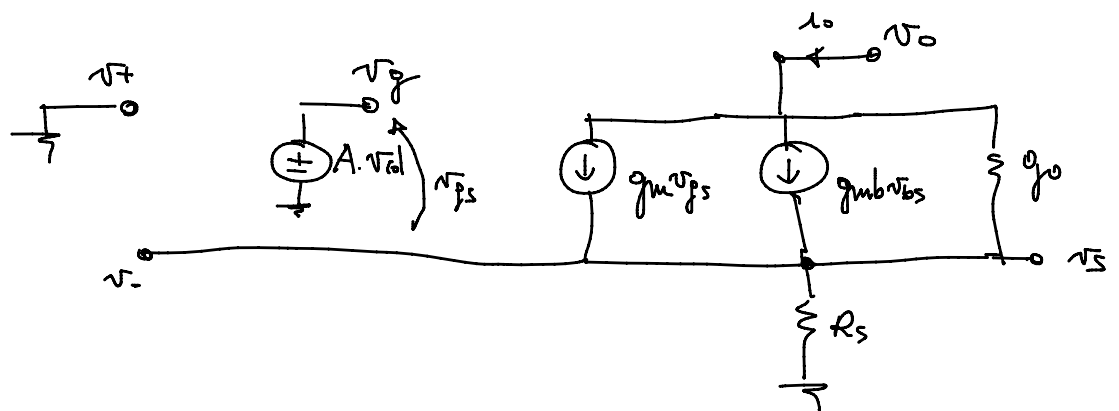
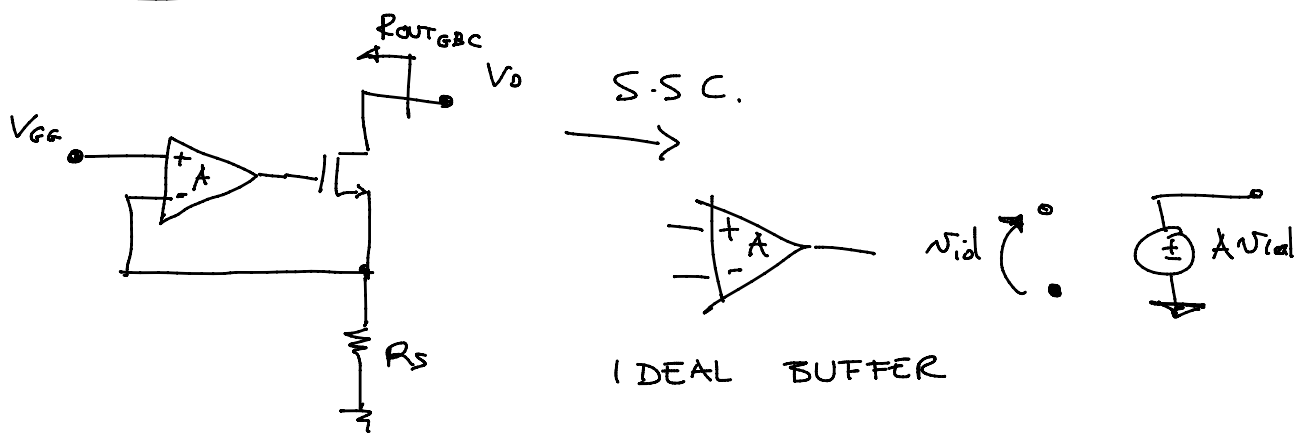


60:00 —

Gain boosted cascode

Notes writing Begins

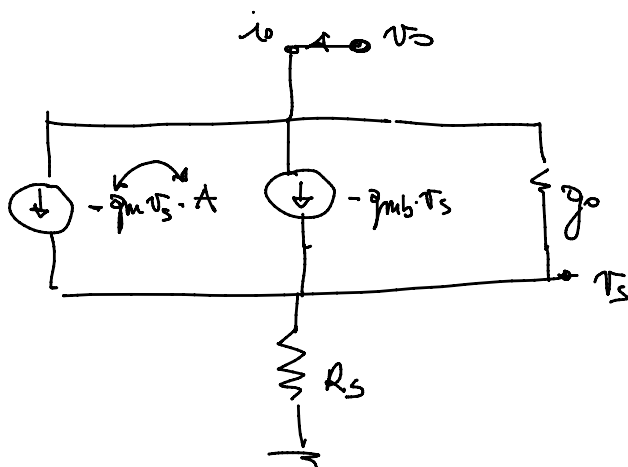
Gain boosted cascode



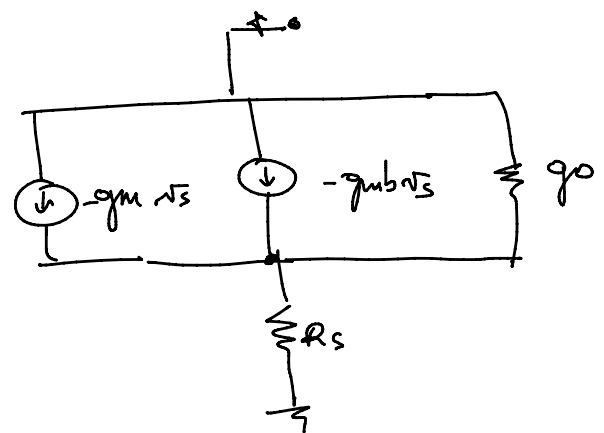
$$v_{id} = v_+ - v_- = v_- = -v_S$$

$$v_{ps} = -A v_S - v_S = -(A+1) \cdot v_S \approx -A v_S$$

Gain boosted

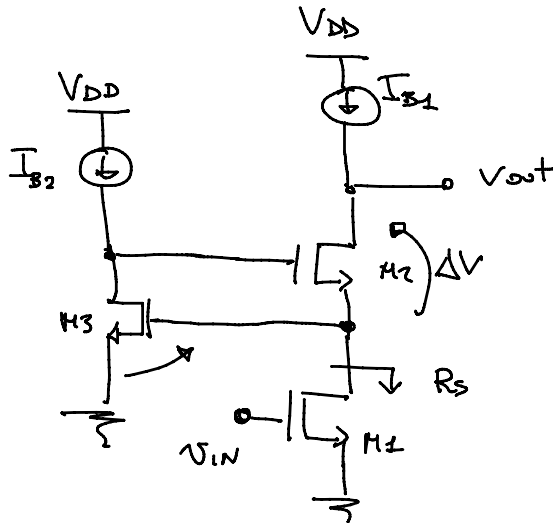


Normal Cascode

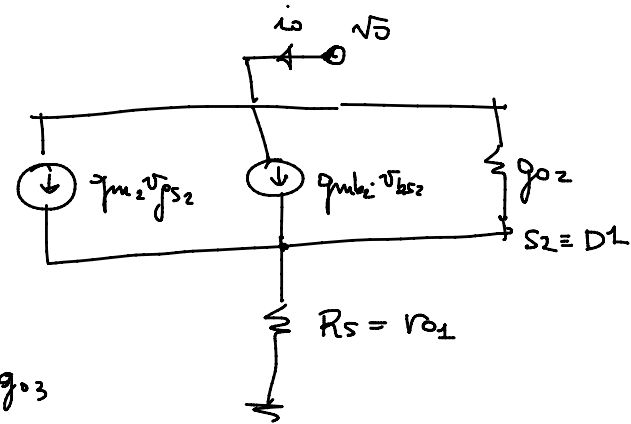
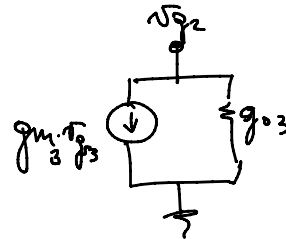


$$R_{o_{GBC}} = g_m \textcircled{A} v_o \cdot R_s$$

$$R_{o_{ase}} = g_m v_o \cdot R_s$$



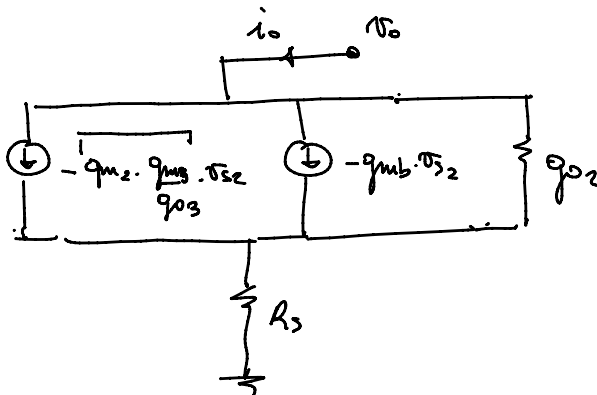
S.S.C.



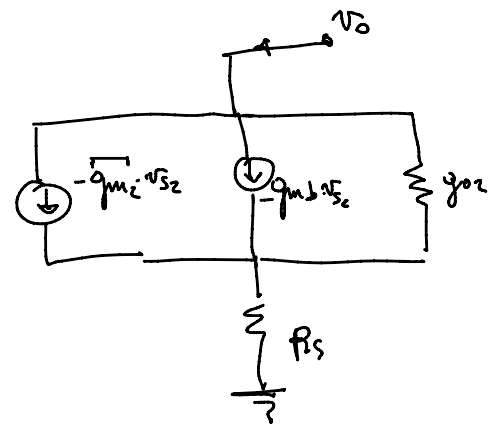
$$\left. \frac{v_o}{i_o} \right|_{v_{in}=0}$$

$$v_{gs3} = v_{s2}$$

$$v_{g2} = -g_{m3} v_{s2} \cdot \frac{1}{g_{o3}} ; \quad \frac{v_{gs2}}{v_{g2} - v_{s2}} = \left(-\frac{g_{m3}}{g_{o3}} - 1 \right) v_{s2} \approx \left[-\frac{g_{m3}}{g_{o3}} \cdot v_{s2} \right]$$



$$R_{out} = \underbrace{g_{m2} \cdot g_{m3} \cdot r_{o3} \cdot r_{o2}}_{\sim 100} \cdot R_s \quad ? \quad 100k\Omega$$



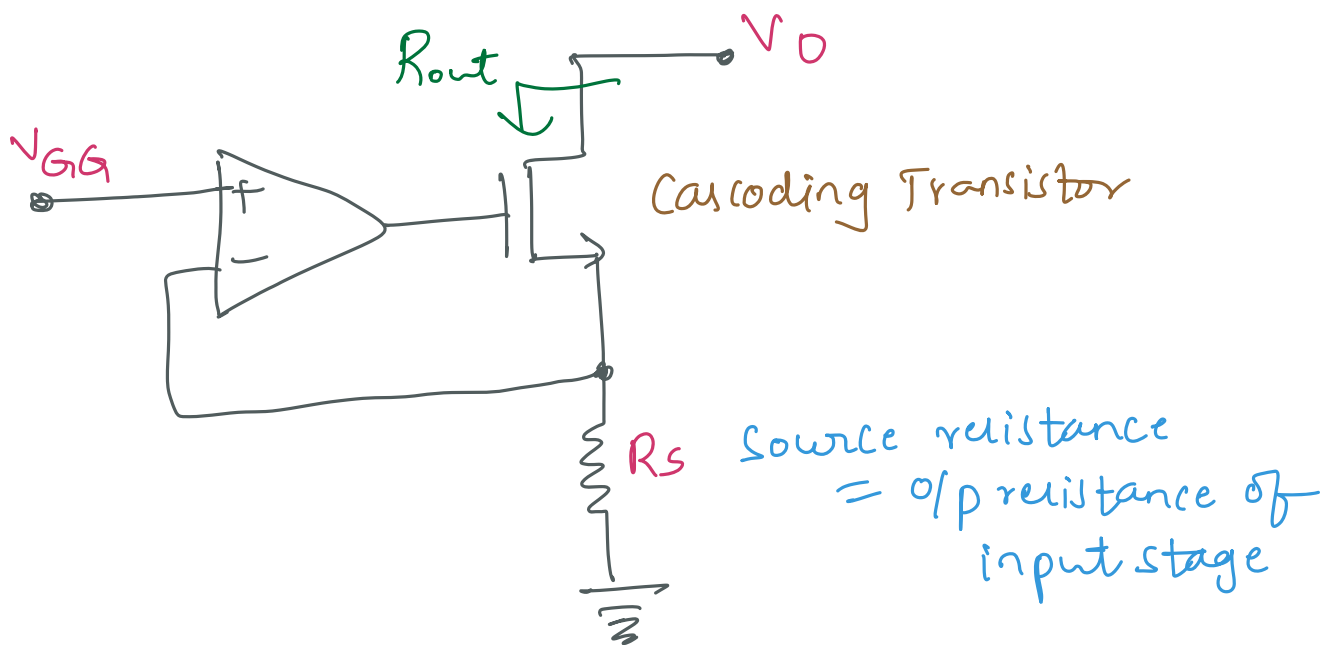
$$R_{out}^+ = \underbrace{g_{m2} \cdot r_{o2}}_{100} \cdot R_s$$

$1 \text{ G}\Omega$

$10 \text{ M}\Omega$

Regulated Cascode

Gain Boosted cascode



* Now we want to determine The o/p resistance of the Gain Boosted cascode.

→ Because High Gain is achieved by having High o/p Resistance at o/p Node.

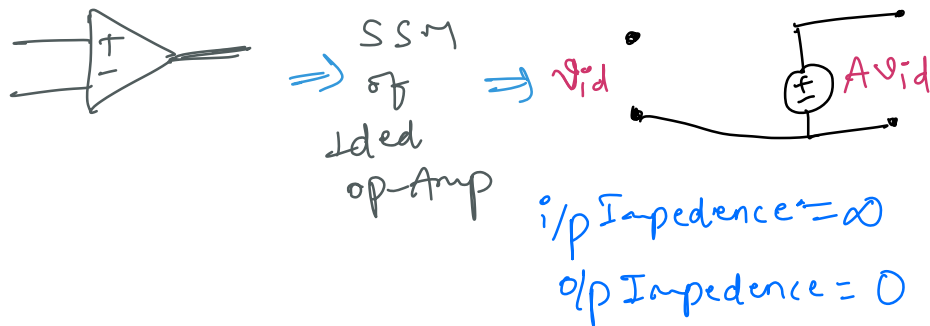
$$A_v = g_m (R_{out_{casc}} || R_L)$$

↓

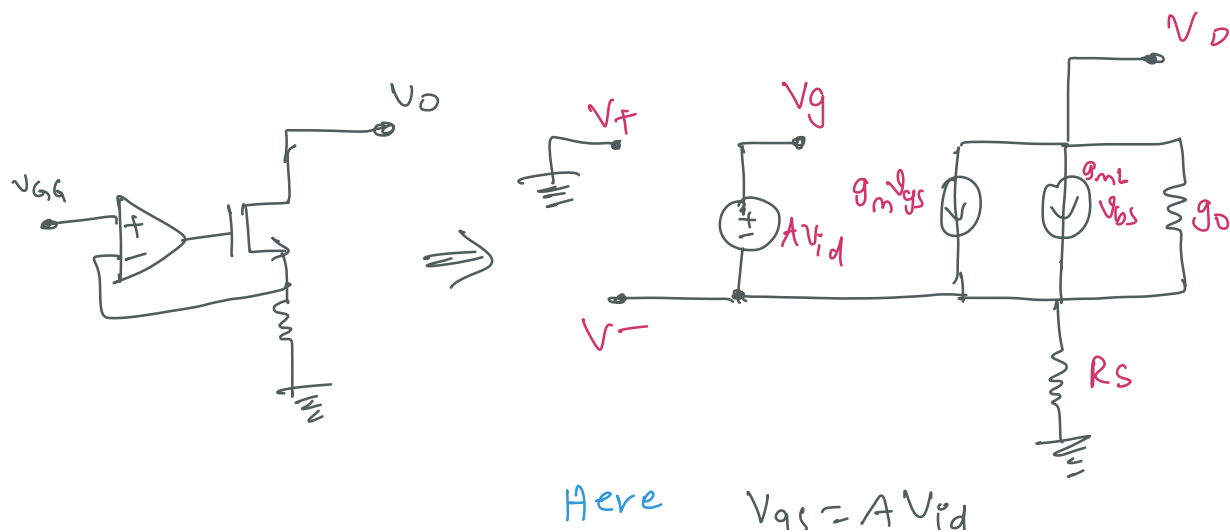
If the o/p resistance of cascode stage is higher then we will have higher Gain.

→ The o/p Resistance of Gain Boosted Cascode Stage is higher than the Normal cascode Stage configuration.

* small signal equivalent of Gain Boosted cascode



plug this in SSM of Gain Boosted Cascode.



$$\Rightarrow V_{gs} = A V_{id}$$

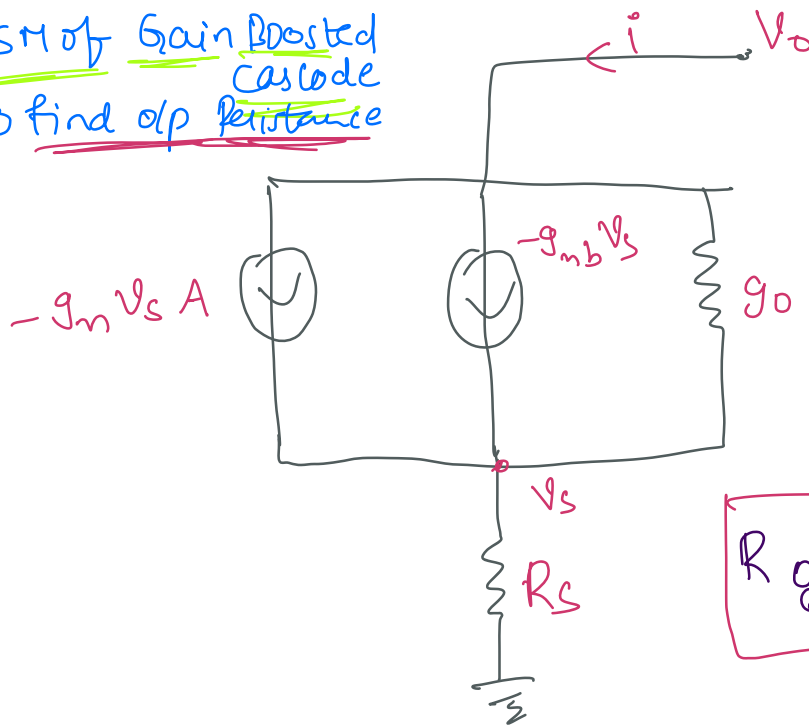
$$V_{id} = V_+ - V_- =$$

$V_+ = 0$ for ssm $\therefore V_+$ is a constant value V_{GG}

$$\Rightarrow V_{id} = -V_- = -V_s$$

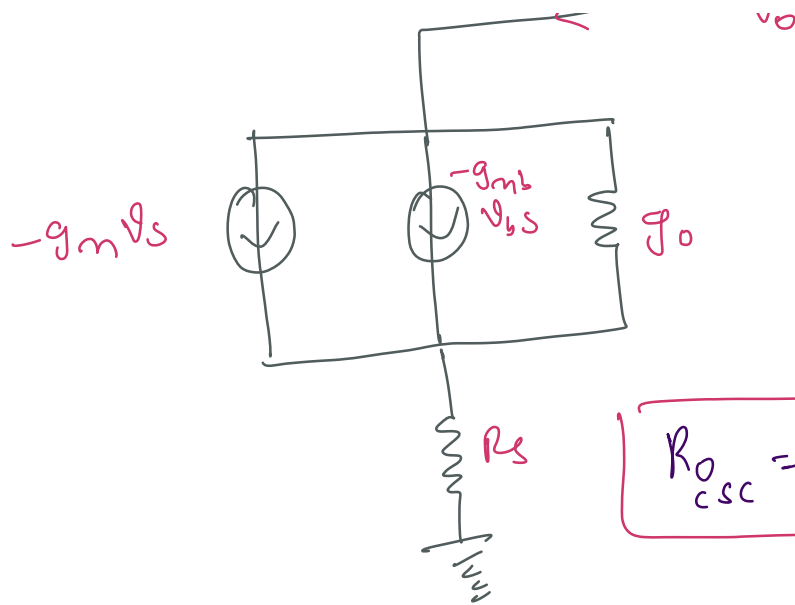
$$\Rightarrow V_{gs} = A(-V_s) = -A V_s$$

SSM of Gain Boosted cascode
to find o/p Resistance

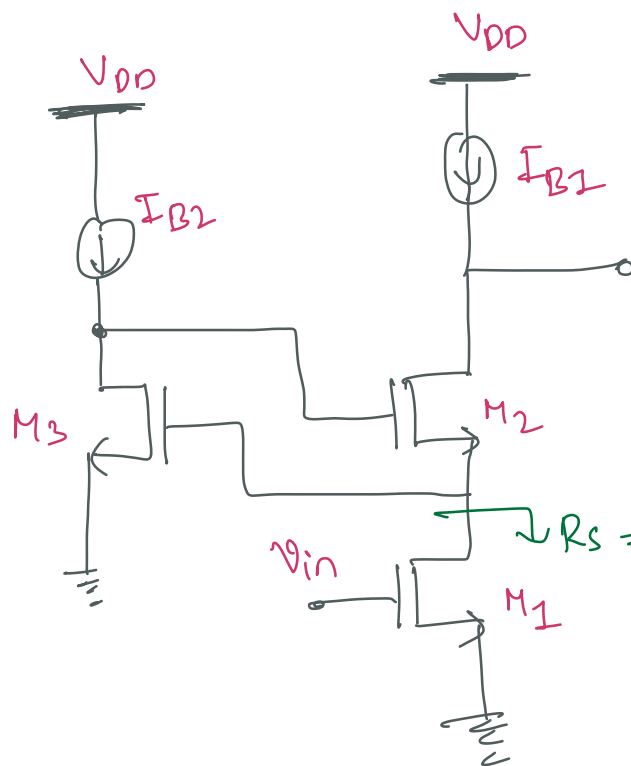


$$R_{o_{GBC}} = g_m A V_o R_s$$

SSM of Normal Cascode (in lecture-9)
to find o/p Resistance page-10



$$R_{o_{csc}} = g_m v_o \cdot R_s$$

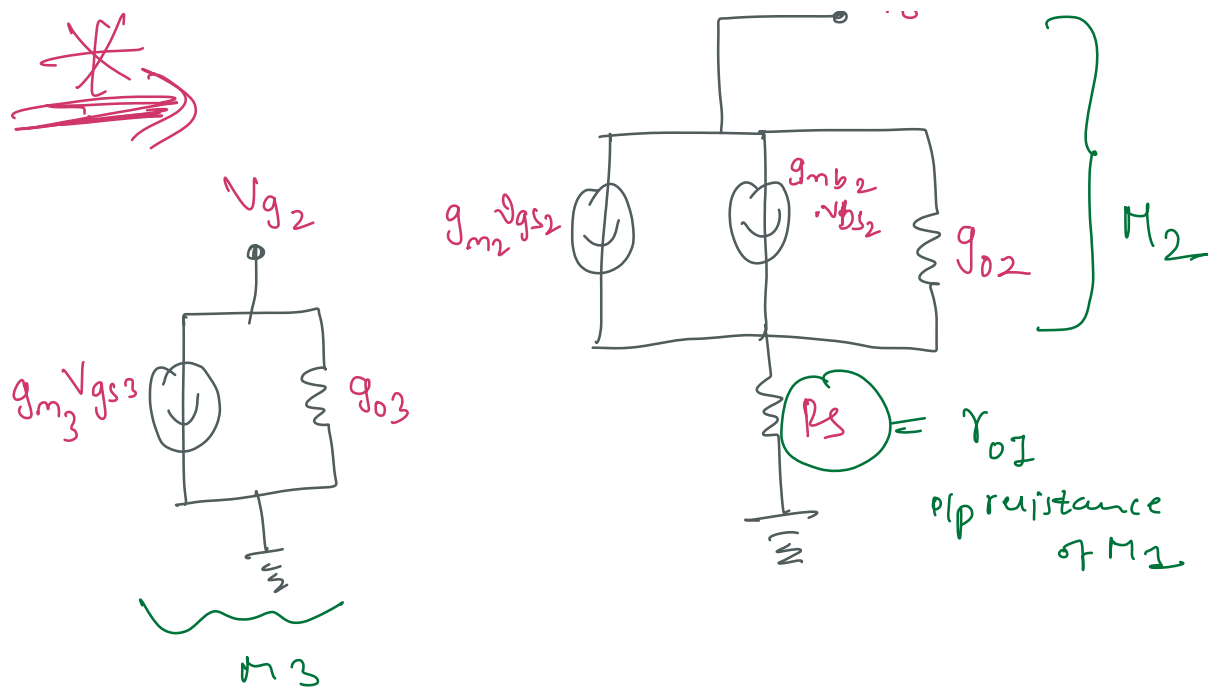


I_{B1}, I_{B2} are
Biasing currents.

$R_s \Rightarrow$ Intrinsic o/p resistance
of M_1

$\Downarrow$ SSC

V_A



we have $V_{gs3} = V_{s2}$ for M_3 Transistor

$$\therefore V_{g2} = V_o \text{ of } M_3$$

$$V_{g2} = -g_{m3} V_{s2} \cdot \frac{1}{g_{o3}}$$

Then

$$V_{gs2} = V_{g2} - V_{s2}$$

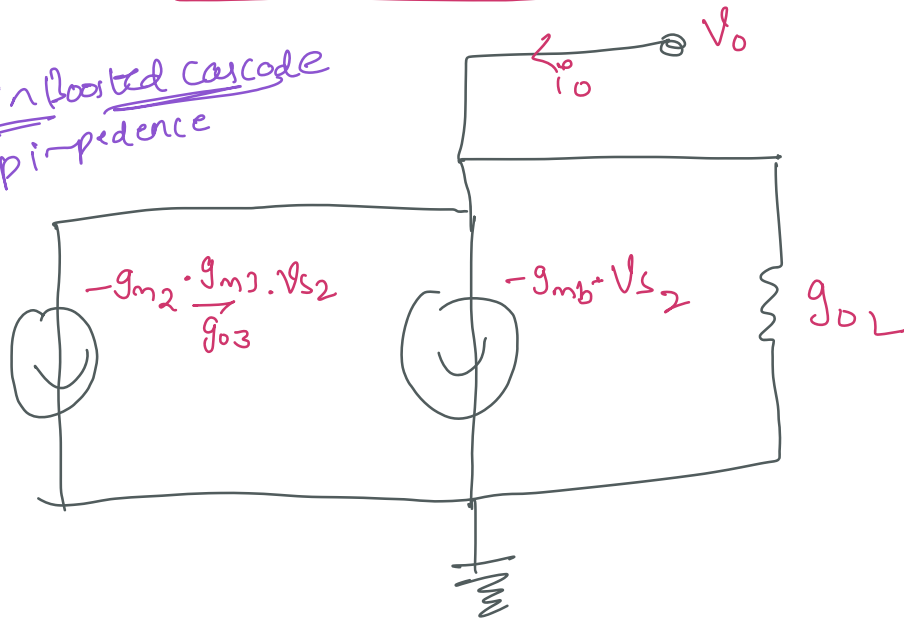
$$= -g_{m3} V_{s2} \cdot \frac{1}{g_{o3}} - V_{s2}$$

$$V_{gs2} = V_{s2} \left(\frac{-g_{m3}}{g_{o3}} - 1 \right)$$

This term is more than 1

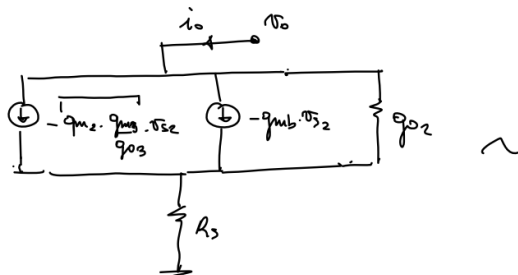
$$\therefore V_{gs2} = -\frac{g_{m3} \cdot V_{s2}}{g_{o3}}$$

* Gain Boosted cascode
output impedance



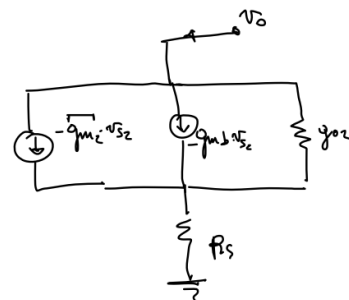
*

Gain Boosted cascode



$$R_{out} = \underbrace{g_{m2} \cdot g_{m3} \cdot V_{s2}}_{\sim 100} \cdot V_{s2} \cdot R_s \quad ? 100k\Omega$$

Normal cascode



$$R_{out} = \underbrace{g_{m2} \cdot V_{s2}}_{\sim 100} \cdot R_s$$

$$R_{out} = \underbrace{g_{m2} \cdot g_{m2} \cdot r_{o3} \cdot r_{o2}}_{10000} \cdot R_S$$

↓

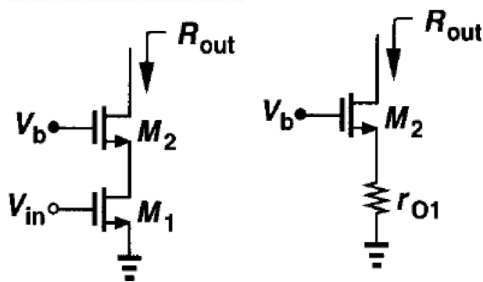
for normal
 R_S was
 multiplied
 by only 100.

For Gain Boosted Cascode
 R_S is multiplied by 10000

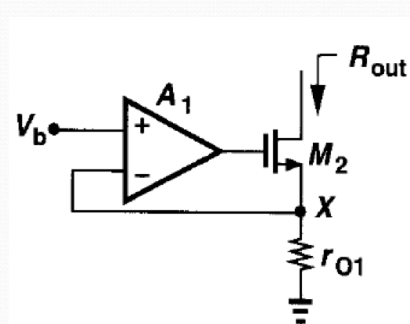
∴ It is called Regulated Cascode.

But, in this case it is difficult to bias a
 Gain Boosted Transistor.

Gain Boosting in Cascode

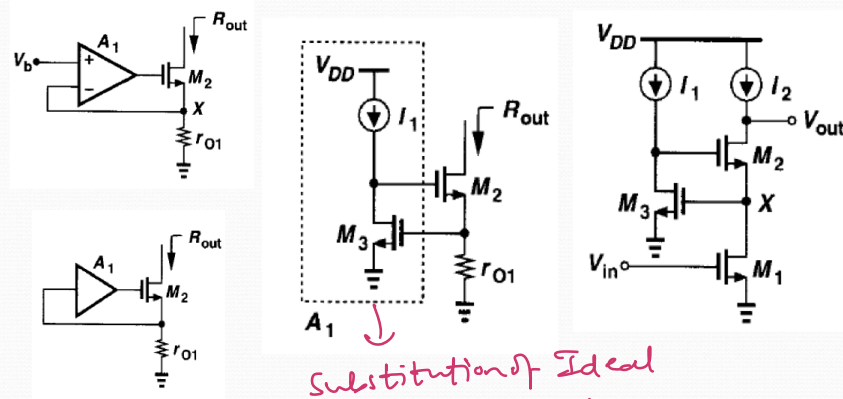


$$R_{out} = g_{m2} r_{o2} r_{o1}$$



$$R_{out} \approx A_1 g_{m2} r_{o2} r_{o1}$$

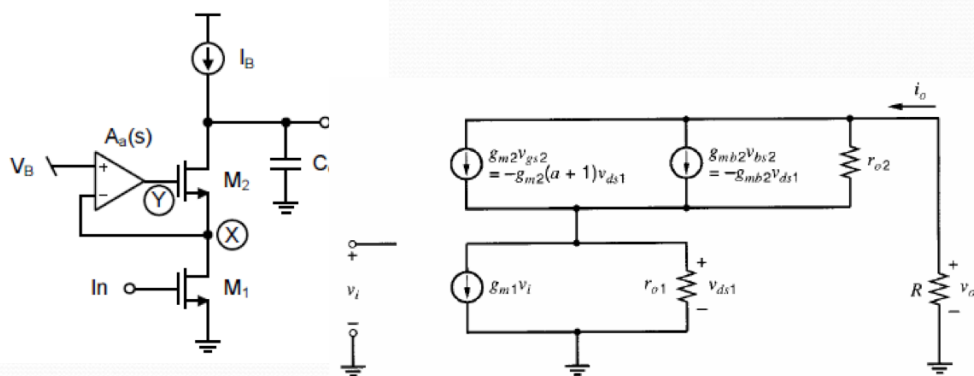
Gain Boosting in Cascode Stage



RZ

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Gain Boosting in Cascode



$$v_{gs2} = v_{g2} - v_{s2} = v_{g2} - v_{ds1} = -(a)v_{ds1} - v_{ds1} = -(a+1)v_{ds1}$$

$$v_{gs2} = -v_{ds1} \text{ in a simple cascode}$$

$$g_{m2} \quad (a+1)g_{m2}$$

GM

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Gain Boosting in Cascode

$$G_m = \left. \frac{i_o}{v_i} \right|_{v_o=0} = g_{m1} \left(1 - \frac{1}{1 + (g_{m2} + g_{mb2}) r_{o1} + \frac{r_{o1}}{r_{o2}}} \right) \approx g_{m1}$$

$$R_o = r_{o1} + r_{o2} + (g_{m2} + g_{mb2}) r_{o1} r_{o2} \approx (g_{m2} + g_{mb2}) r_{o1} r_{o2}$$

Cascode

$$G_m = g_{m1} \left(1 - \frac{1}{1 + [g_{m2} (a + 1) + g_{mb2}] r_{o1} + \frac{r_{o1}}{r_{o2}}} \right) \approx g_{m1}$$

$$R_o = r_{o1} + r_{o2} + [g_{m2} (a + 1) + g_{mb2}] r_{o1} r_{o2} \approx [g_{m2} (a + 1) + g_{mb2}] r_{o1} r_{o2}$$

GainB.
Cascode

CASCODE vs. CASCADE?

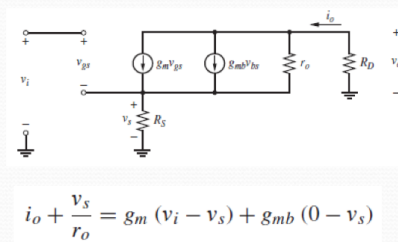
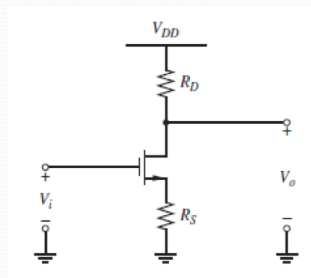
GM

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MOS stage with a load on both Drain and Source

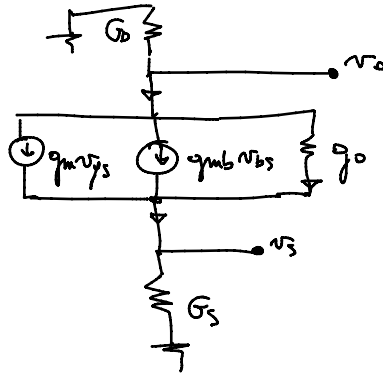
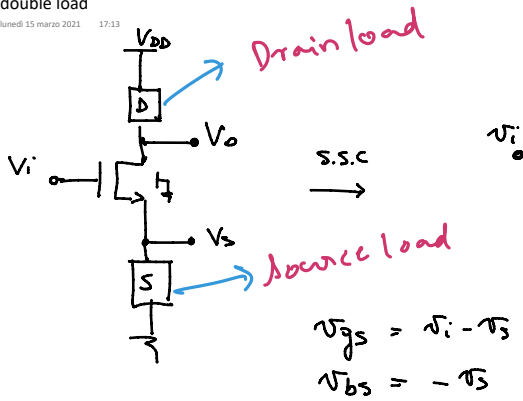
MOS stage with a load on both drain and source



$$G_m = \frac{i_o}{v_i} = \frac{g_m}{1 + (g_m + g_{mb}) R_S + \frac{R_S}{r_o}} \approx \frac{g_m}{1 + (g_m + g_{mb}) R_S}$$

$$R_o = \frac{v_t}{i_t} = R_S + r_o [1 + (g_m + g_{mb}) R_S]$$

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$$G_S = \frac{1}{R_S}$$

$$G_D = \frac{1}{R_D}$$

$$g_m v_{gs} + g_{mb} v_{gs} + g_o (v_o - v_s) = G_S v_s$$

$$G_D (0 - v_o) = G_S v_s \Rightarrow v_s = -\frac{G_D}{G_S} v_o$$

$$g_m v_i = G_T v_s - g_o v_o + G_S v_s$$

$$g_m v_i = -(G_T + G_S) \frac{G_D}{G_S} v_o - g_o v_o = -v_o \left(\frac{(G_T + G_S) G_D}{G_S} + g_o \right)$$

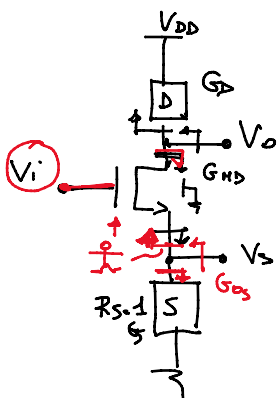
$$\frac{v_o}{v_i} = \frac{-g_m}{G_D + g_o + \frac{G_T G_D}{G_S}} = -\frac{g_m R_D}{1 + g_o R_D + \frac{G_T}{G_S}} = -\frac{g_m R_D}{1 + g_o R_D + G_T R_S}$$

$$v_o = -\frac{G_S}{G_D} v_s$$

$$g_m v_i = (G_T + G_S) v_s + g_o \frac{G_S}{G_D} v_s$$

$$\frac{v_s}{v_i} = \frac{g_m}{G_T + G_S + g_o \frac{G_S}{G_D}} \xrightarrow{G_D \rightarrow \infty} = \frac{g_m}{G_T + G_S} \quad \text{C.D.}$$

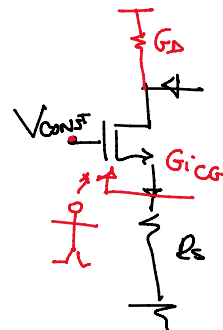
$$\frac{v_o}{v_i} = \frac{-g_m}{G_D + g_o + \frac{G_T G_D}{G_S}} \xrightarrow{G_S \rightarrow \infty} = -\frac{g_m}{G_D + g_o} \quad \text{C.S.}$$



$$G_{o_d} = G_D + G_{MD} \quad \text{C.D.}$$

$$G_{out} = G_D + G_{MD} \quad \text{C.S.}$$

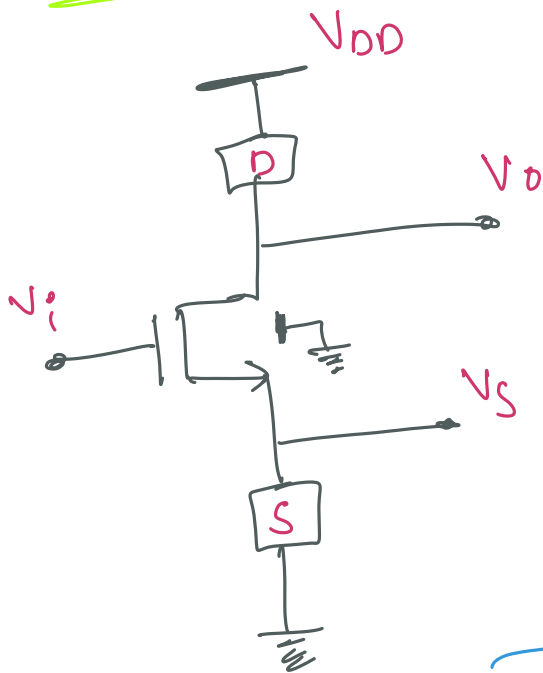
$$G_{os} = G_S + G_{icg}$$



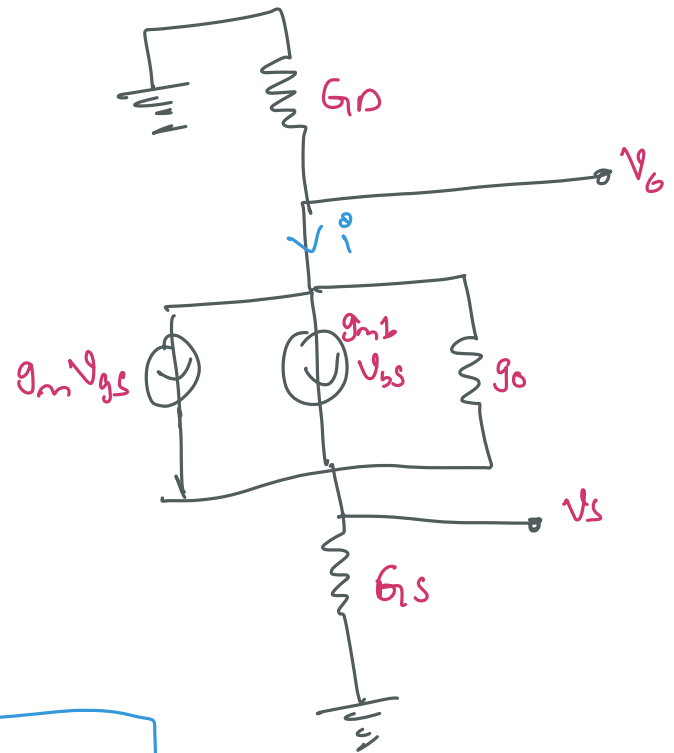
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$$G_{os} \Big|_{V_{in}=0} = G_S + \underline{\underline{G_{ice}}}$$

Double load



SSC



$$\begin{aligned} V_{gs} &= V_i - V_s \\ V_{bs} &= 0 - V_s \end{aligned}$$

The current flowing through G_S = current flowing through SSC of NMOS

$$\Rightarrow V_s G_S = g_m V_{gs} + g_{mb} V_{bs} + g_o (V_o - V_s)$$

$$\Rightarrow V_s G_S = g_m (V_i - V_s) + g_{mb} (-V_s) + g_o (V_o - V_s) \quad \text{--- (1)}$$

The current flowing through The G_D resistor is in opposite direction to what flows through G_S .

$$\therefore V_s G_S = -V_o G_D$$

$$\Rightarrow \boxed{V_S = -V_O \left(\frac{G_D}{G_S} \right)} \quad \text{--- (2)}$$

from (1)

$$V_S G_S = g_m V_1 - g_m V_S - g_{m_b} V_S + g_o V_O - g_o V_S$$

$$V_S G_S \Rightarrow g_m V_1 - V_S (g_m + g_{m_b} + g_o) + g_o V_O$$

$$\Rightarrow V_S G_S \Rightarrow g_m V_1 - V_S G_T + g_o V_O$$

$$\Rightarrow \boxed{g_m V_1 = V_S G_S + G_T V_S - g_o V_O}$$

$$\Rightarrow g_m V_1 = (G_S + G_T) V_S - g_o V_O$$

$$\Rightarrow g_m V_1 = (G_S + G_T) \left(-V_O \cdot \frac{G_D}{G_S} \right) - g_o V_O$$

$$\Rightarrow g_m V_1 = \left(- (G_S + G_T) \left(\frac{G_D}{G_S} \right) - g_o \right) V_O$$

$$\Rightarrow \boxed{\frac{V_O}{V_1} = \frac{-g_m}{(G_S + G_T) \left(\frac{G_D}{G_S} \right) + g_o}}$$

$$\Rightarrow \boxed{\frac{V_O}{V_1} = \frac{-g_m R_D}{G_o}}$$

$$V_1 = \frac{1}{1 + g_o R_D + \frac{G_T}{G_S}}$$

$$\frac{V_o}{V_1} = \frac{-g_m R_D}{1 + g_o R_D + G_T R_S}$$



$$V_o = -\frac{G_S}{G_D} \cdot V_S$$

$$g_m V_i = (G_T + G_S) V_S + g_o \frac{G_S}{G_D} \cdot V_S$$

$$\frac{V_S}{V_i} = \frac{g_m}{G_T + G_S + g_o \frac{G_S}{G_D}} \xrightarrow{G_D \rightarrow \infty} = \frac{g_m}{G_T + G_S} \quad \text{C.D.}$$

Relationship b/w V_S & V_i

→ we are doing all these calculations because double load configuration will be Ground Bias for differential amplifier

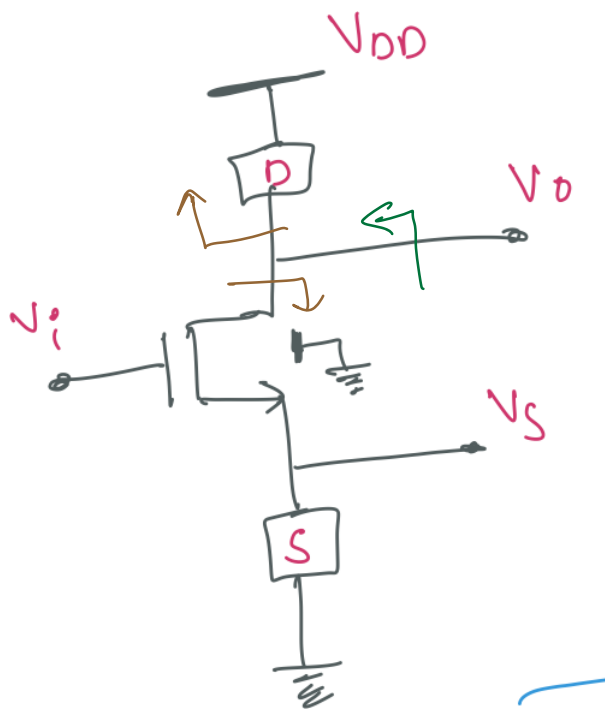
ద్రవ్యం పల్లె పుష్కల

Relationship b/w Drain and Gate terminal voltage

$$\frac{v_o}{v_i} = \frac{-g_m}{G_D + g_o + \frac{G_T \cdot G_D}{G_S}} \xrightarrow{G_S \rightarrow \infty} = - \frac{g_m}{G_D + g_o} \quad \text{CS}$$

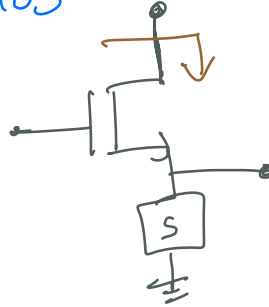
i.e. $R_S = 0$
 which means source of the
 NMOS is shorted to GND
 which is a normal
 Common source stage -

→ we have calculated Gains $\frac{v_o}{v_i}$, $\frac{v_s}{v_i}$

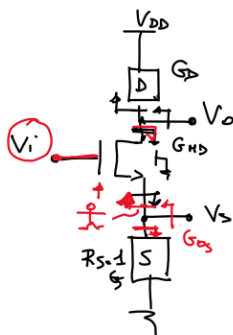


But calculation of
 o/p Conductance is
 also important.

$\Rightarrow$ o/p conductance seen through node V_o is equal to parallel of the conductance we see towards  and into the NMOS



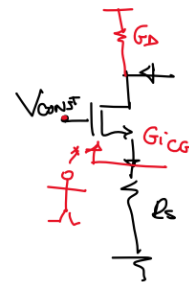
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$$G_{oD} \Big|_{v_{in}=0} = G_D + G_{MD}$$

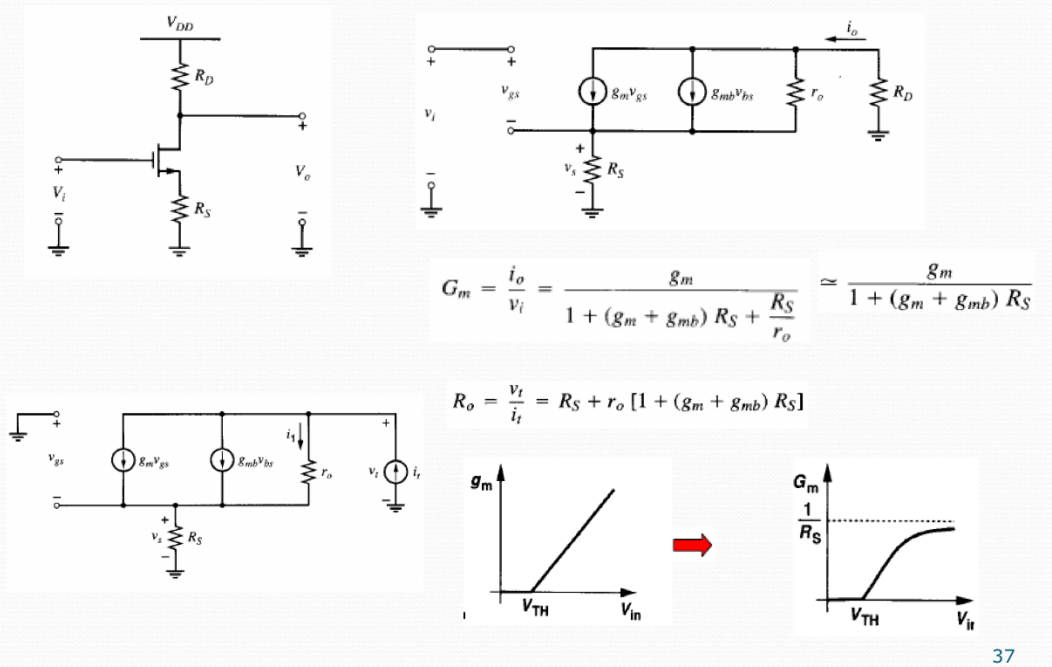


$$G_{oS} \Big|_{\cdot} = G_S + G_{icg}$$



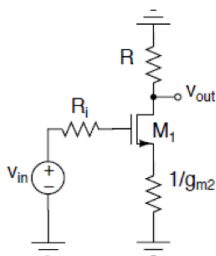
$$G_{oS} \Big|_{v_{in}=0} = G_S + G_{icg}$$

Source Degeneration



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Toward a New Circuit



- If R_S is small enough (e.g. $1/g_{m2}$) the global G_m halves the one of the CS

$$G_m = \frac{g_{m1}}{1 + g_{m1} \times \frac{1}{g_{m2}}} = \frac{g_m}{2}$$

$$\Rightarrow a_{v0} = G_m \times R = -\frac{1}{2} g_m R$$

- It looks like we are losing half of the gain.
- Is there any way we can retrieve that other half?

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